Anisotropic Thin Shell Around a $3\,M_\odot$ Core

Static Equilibrium, Radial Stability, Vacuum Limit, Thickness Constraints, Equation of State, Constant-Sound-Speed Closure and Finite-Thickness Test

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Note on this edition

This monograph collects Parts I–IV of the series in a single volume with a common bibliography and common equation numbering (equations are numbered within each Part, e.g. (I.11)). Cross-references between the Parts have been updated accordingly. Where the Parts disagreed, the following harmonisations were made: (1) the lower bound on the shell density required for $l/a \leq 0.1$ is stated uniformly as 11–33 ρ_{nuc} (Table I.2); (2) the $\approx 6\%$ agreement of Part I is stated as referring to the two core models of Table I.1, with the 54-model scan of Part II quoted alongside; (3) the DEC window of Part I is identified as the thin-shell (mean) DEC, with a pointer to the local analysis of Part IV; (4) rows of Table I.2 above the causal limit $\mathcal{C}_+ \leq 0.844$ are marked as illustrative; (5) the several estimates of the shell density at $\mathcal{C}_+ = 0.785$ (13, 15.5 and 19 ρ_{nuc}) are listed together in Part III; (6) the distinction between the layer model of Part IV and the core–shell configuration of Part I is stated explicitly. No derivation or numerical result was changed.

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PART I

Anisotropic Thin Shell Around a $3\,M_\odot$ Core: Static Equilibrium, Radial Stability, Vacuum Limit, and Thickness Constraints

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Abstract. We present a model investigating whether collapsing matter of mass around $3\,M_\odot$ can avoid a singularity by forming a rigid thin shell with anisotropic stresses surrounding a regular fluid core. The interior core is described by an anisotropic Tolman–Oppenheimer–Volkoff (TOV) system with closure $p_r = \rho/3$ and pressure anisotropy $\Delta = p_t - p_r = \alpha\rho r^2/a^2$, while the boundary is governed by the Darmois–Israel junction conditions. Static configurations adhering to the dominant energy condition (DEC) in its thin-shell (mean) form, $\sigma \geq |P|$, exist within the external compactness window $\mathcal{C}_+ = 2GM/(c^2 a) \in [0.785, 0.945]$ for $\alpha \leq 0.5$ and central densities $\rho_c = (1\text{–}4)\,\rho_{\rm nuc}$; requiring in addition a causal shell stiffness $\eta_0 \leq 1$ restricts stable configurations to $\mathcal{C}_+ \leq 0.844$. From a linear radial stability analysis explicitly incorporating matter flux across the shell, we determine the critical shell stiffness $\eta_{\rm crit}(\mathcal{C}_+)$ required for stability ($\eta_0 = dP/d\sigma > \eta_{\rm crit}$). We show analytically that the sign of the flux term originates strictly from the boundary normal orientation. Furthermore, for a vacuum interior ($m_- = 0$), we derive a closed-form analytic expression for the critical stiffness, $\eta_{\rm vac} = (1-y)(1+2y+3y^2)/[4y^2(1+3y)]$ with $y = \sqrt{1-\mathcal{C}_+}$, which reproduces the numerical results for the two core models studied here within $\approx 6\%$ (a scan of 54 core models in Part II gives -3.8% to $+9.7\%$ for $0.785 \leq \mathcal{C}_+ \leq 0.845$), demonstrating that stability is primarily governed by exterior geometry. Finally, estimating shell thickness yields $l/a \approx 1.1\text{–}3.3$ at nuclear density, implying that self-consistency of the thin-shell approximation ($l/a \leq 0.1$) requires extreme densities $\rho_s \geq 11\text{–}33\,\rho_{\rm nuc}$. The local form of the DEC inside a layer of finite thickness is tested in Part IV.

Keywords: thin shell, pressure anisotropy, Israel junction conditions, radial stability, compact objects.

I.1 Introduction and Hypothesis

Classical gravitational collapse beyond the maximum mass limit of compact stars leads inevitably to a singularity within general relativity. The hypothesis explored here is that under extreme density conditions, matter generates pressure anisotropy, allowing a rigid thin shell to halt collapse prior to singularity formation. Physically, the shell acts as a tensioned membrane balancing internal pressure from the core against the gravitational attraction of the total mass M.

Hypothesis. Fundamental interactions build matter in a sequence: the Higgs field gives quarks a small part of their mass; the strong interaction binds them into nucleons and generates most of the remaining mass; the weak interaction links neutrons and protons; electromagnetism then builds atoms, molecules and chemistry. The hypothesis is that gravitational compression runs this sequence in reverse. Compression first destroys atoms (electron-degenerate matter), then converts protons and electrons into neutrons (a neutron star), and beyond the neutron-star stage continues to undo the structure built by the interactions: quarks are released from hadronic confinement, the strong binding loses its power, and, in the limit, the mass-generating mechanism itself is switched off. The thin-shell model below does not derive this sequence; it tests whether, for the resulting ultra-dense matter, a static, stable and physically admissible configuration with pressure anisotropy can exist at $M \sim 3\,M_\odot$.

I.1.1 Microphysical Intuition and Scales

A key motivation stems from the characteristic scale $R_i = GMm_0/X_i$, where $m_0 \approx 0.94\,\mathrm{GeV}/c^2$ and X_i represents energy scales of the Standard Model. For $M = 2.8\,M_\odot$, these correspond to 48.35 m ($W^\pm$), 42.62 m (Z^0), 31.06 m (Higgs), and 15.8 m (VEV). These lengths lie deep inside the Schwarzschild radius ($R_s = 8.27\,\mathrm{km}$), while the gravitational potential energy of a nucleon within the regular region satisfies $Gmm_0/r < m_0c^2/2 \approx 0.47\,\mathrm{GeV}$, remaining below X_i. Consequently, R_i does not represent the physical shell radius, but rather serves as a heuristic scale indicator. Here, we address the fundamental geometric and dynamic question: does a static, stable, and physically admissible thin-shell configuration exist for $M \sim 3\,M_\odot$?

I.1.2 Physical Picture: The Idea Behind the Hypothesis

The standard description of the end state of a compact object is set by the TOV equation with a nuclear equation of state [5]: beyond the critical mass, a collapsing medium proceeds toward a gravitational singularity. The working hypothesis behind this paper is that, before the matter reaches the singularity, the extreme densities may activate microphysical thresholds associated with the energy scales X_i of the Standard Model, and that matter at this stage no longer behaves as an ideal isotropic fluid.

Two points follow from the geometry and are stated here so that the picture is not overstated. First, the scales R_i lie deep inside the trapped region of the exterior Schwarzschild metric (for $M = 2.8\,M_\odot$ and $R_W \approx 48.35\,\mathrm{m}$ one finds $2GM/(c^2R_W) \approx 171$), so they cannot be the radius of a static shell; a static shell requires $\mathcal{C}_+ < 1$, i.e. $a > R_s$. The role of a microphysical activation indicator would have to be played by the local compression of matter, i.e. the local energy density, rather than by the gravitational potential alone; the precise form of this criterion remains an open problem. Second, for compactness close to unity ($\mathcal{C}_+ \approx 0.8$–$0.95$, above the Buchdahl bound $2M/R \leq 8/9$ for an isotropic perfect fluid [6]) an isotropic fluid cannot provide static support, so the matter must develop pressure anisotropy [7], $p_t \neq p_r$. The positive anisotropic term in Eq. (I.4) generates an outward force that counteracts gravitational collapse, and since $p_r = \rho/3$ does not vanish at the boundary, the core ends at a sharp phase interface, which is modelled as a thin shell.

The following ingredients are natural extensions of this picture but are not results of the present paper: rotation and frame dragging in the slow-rotation formalism [8]; a nonlinear electroweak gauge sector carrying a conserved topological (Chern–Simons) charge, which by itself fixes no length scale and would require an additional energetic condition; and surface

currents supporting magnetar-scale fields. They are listed here as directions for further work.

I.2 Model Formulation and Boundary Junction

Geometric units $G = c = 1$ are adopted throughout ($GM_\odot/c^2 = 1476.625\,\mathrm{m}$).

I.2.1 Anisotropic Core

The interior region ($r \le a$) is described by the metric $ds^2 = -e^{2\nu}dt^2 + dr^2/(1-2m/r) + r^2 d\Omega^2$ with stress-energy tensor $T^\mu{}_\nu = \mathrm{diag}(-\rho, p_r, p_t, p_t)$. The anisotropic TOV structural equations [7, 9] are:

$$\frac{dm}{dr} = 4\pi r^2 \rho, \tag{I.1}$$

$$\frac{dp_r}{dr} = -\frac{(\rho + p_r)(m + 4\pi r^3 p_r)}{r(r-2m)} + \frac{2(p_t - p_r)}{r}. \tag{I.2}$$

We adopt the equation of state closure $p_r = \rho/3$ alongside a quadratic spatial anisotropy:

$$\Delta(r) = p_t - p_r = \alpha\, \rho(r)\, \frac{r^2}{a^2}, \tag{I.3}$$

which leads to the density gradient equation:

$$\frac{d\rho}{dr} = -\frac{4\rho\left(m + \frac{4\pi}{3} r^3 \rho\right)}{r(r-2m)} + \frac{6\Delta}{r}. \tag{I.4}$$

With boundary conditions $m(0) = 0$ and $\rho(0) = \rho_c$, satisfying the Dominant Energy Condition (DEC) in the interior ($\rho \ge |p_t|$) requires $1/3 + \alpha r^2/a^2 \le 1$, yielding $\alpha \le 2/3$. Because radial pressure $p_r = \rho/3$ does not vanish at $r = a$, the boundary is a sharp phase interface.

I.2.2 Shell Junction Conditions and Flux Sign Derivation

At the surface $r = a$, we define compactness parameters $\mathcal{C}_+ = 2M/a$ (exterior) and $\mathcal{C}_- = 2m_-(a)/a$ (interior), with $q_- = 4\pi a^2 \rho_-(a)$. The static Darmois–Israel junction conditions [10] yield surface density σ and surface pressure P:

$$4\pi a \sigma = \sqrt{1-\mathcal{C}_-} - \sqrt{1-\mathcal{C}_+}, \tag{I.5}$$

$$8\pi a P = \frac{\mathcal{C}_+}{2\sqrt{1-\mathcal{C}_+}} - \frac{m_-/a + q_-/3}{\sqrt{1-\mathcal{C}_-}} + \sqrt{1-\mathcal{C}_+} - \sqrt{1-\mathcal{C}_-}. \tag{I.6}$$

The term $q_-/3 = 4\pi a^2 p_r^-(a)$ arises from $K^\tau{}_\tau = \sqrt{f_-}\,\nu'$ on the inner side. For a vacuum interior ($m_- = q_- = 0$), these equations reduce to standard vacuum thin-shell formulas. Positivity of surface mass density ($\sigma > 0$) requires $m_- < M$. DEC for the shell demands $\sigma \ge |P|$.

Derivation of the Flux Term Sign

The energy-momentum conservation across the matching hypersurface is governed by:

$$S^{ab}{}_{|b} = -\left[T_{\mu\nu} e^\mu_a n^\nu\right], \tag{I.7}$$

where n^μ is the unit normal pointing outward from interior to exterior, and $[X] = X_+ - X_-$. Projecting onto the shell velocity u^a using $S^{ab} = \sigma u^a u^b + P(g^{ab} + u^a u^b)$ and divergence $u^a{}_{|a} = 2\dot{a}/a$, we obtain $u_a S^{ab}{}_{|b} = -[\dot{\sigma} + 2(\sigma + P)\dot{a}/a]$. As the exterior is vacuum, $[T_{\mu\nu}u^\mu n^\nu] = -T^-_{\mu\nu}u^\mu n^\nu$, so:

$$\dot{\sigma} + \frac{2}{a}(\sigma + P)\dot{a} = -T^-_{\mu\nu}u^\mu n^\nu. \tag{I.8}$$

In the interior, $u^\mu = (Ae^{-\nu}/\sqrt{f_-}, \dot{a}, 0, 0)$, $n^\mu = (-e^\nu \dot{a}/\sqrt{f_-}, A/f_-, 0, 0)$, fluid velocity $U^\mu = (e^{-\nu}, 0, 0, 0)$, and radial vector $\hat{r}^\mu = (0, \sqrt{f_-}, 0, 0)$, satisfying $u \cdot n = 0$ and $n \cdot n = 1$. Evaluating the energy tensor $T^-_{\mu\nu} = (\rho + p_t)U_\mu U_\nu + p_t g_{\mu\nu} + (p_r - p_t)\hat{r}_\mu \hat{r}_\nu$ yields:

$$T^-_{\mu\nu}u^\mu n^\nu = (\rho + p_r)\frac{A\dot{a}}{f_-}. \tag{I.9}$$

Dividing by $\dot{a}$ yields the shell energy evolution equation:

$$\frac{d\sigma}{da} = -\frac{2}{a}(\sigma + P) - \frac{(\rho + p_r)A}{f_-}. \tag{I.10}$$

Thus, the negative sign preceding the flux term is uniquely dictated by the orientation of the normal vector n^μ and convention $[X] = X_+ - X_-$, requiring no ad hoc tuning for equilibrium.

I.3 Numerical Method and Static Equilibrium

The interior system was integrated using the high-order DOP853 integrator with relative tolerance 10^{-10} from $r_0 = 1\,\mathrm{mm}$ to $r = a$. Nuclear saturation density is set to $\rho_{\mathrm{nuc}} = 2.7 \times 10^{17}\ \mathrm{kg/m^3}$.

Static shell configurations satisfying DEC ($|P/\sigma| \leq 1$) exist across the compactness range $\mathcal{C}_+ \in [0.785, 0.945]$ for core parameters $\alpha \leq 0.5$, central densities $\rho_c = (1\text{–}4)\,\rho_{\mathrm{nuc}}$, and masses $M \in [2.8, 3.2]\,M_\odot$. The core typically accounts for 0.4–2.5 $M_\odot$, while the shell carries 0.7–2.6 $M_\odot$. For every core in this parameter range the DEC holds throughout this window (checked numerically). The exact edges depend on the core: the upper edge $\mathcal{C}_+ \approx 0.945$ is reached for the most extreme parameters in the range (3.2 $M_\odot$, 4ρ_{nuc}, $\alpha = 0.5$), and the lower edge 0.785 is a conservative choice, since the DEC also holds down to $\mathcal{C}_+ \approx 0.74$ for all tested cores.

Note for the monograph edition. The condition $\sigma \geq |P|$ used here is the thin-shell DEC, which tests the mean ratio P/σ. The local DEC inside a layer of finite thickness is a stronger condition: for tangential-only matter it holds only for $\mathcal{C}_+ \leq 0.8$, and for $0.8 < \mathcal{C}_+ \leq 0.8437$ a small radial pressure is needed (Part IV). The window quoted above should therefore be read as the window of the thin-shell DEC.

I.4 Radial Stability and Vacuum Boundary Limit

Applying linear stability analysis [11] under a static core assumption (the interior fluid is static and its profile is not perturbed, so that $m(a)$ and $\rho(a)$ follow the equilibrium TOV profile as the shell moves), the motion of the shell is governed by an effective potential $\dot{a}^2 + V(a) = 0$, where $V(a) = f_-(a) - A^2$ and $A = \frac{1}{2}\left[s + \frac{2(M-m)}{as}\right]$ with $s = 4\pi a\sigma$. A linear equation of state is

assumed near equilibrium: $P = P_0 + \eta_0(\sigma - \sigma_0)$, where $\eta_0 = dP/d\sigma|_{a_0}$. Stability ($V''(a_0) > 0$) requires $\eta_0 > \eta_{\rm crit}$.

I.4.1 Vacuum Limit Closed-Form Expression

For a vacuum interior ($m_- = 0$), the flux vanishes and $s' = -(s + \Pi)/a$, where $\Pi = 8\pi a P$. Static equilibrium $V' = 0$ yields $s_0 = 1 - y$ and $\Pi_0 = (1-y)^2/(2y)$, where $y = \sqrt{1 - \mathcal{C}_+}$. Solving $A'' = 0$ algebraically yields an exact closed formula for the critical stiffness:

$$\eta_{\rm vac}(\mathcal{C}_+) = \frac{(1-y)(1+2y+3y^2)}{4y^2(1+3y)}, \qquad y = \sqrt{1-\mathcal{C}_+}. \tag{I.11}$$

In the limits:

$$\eta_{\rm vac} \approx \frac{3}{16}\mathcal{C}_+ \ \ (\mathcal{C}_+ \to 0), \qquad \eta_{\rm vac} \approx \frac{1}{4(1-\mathcal{C}_+)} \ \ (\mathcal{C}_+ \to 1). \tag{I.12}$$

Imposing the causal sound speed limit $\eta_0 \le 1$ gives $\mathcal{C}_+ \le 0.844$, corresponding to radii $a \ge 9.80\,\text{km}$ ($2.8\,M_\odot$) and $a \ge 11.20\,\text{km}$ ($3.2\,M_\odot$).

Table I.1. Critical shell stiffness $\eta_{\rm crit}$ comparing TOV cores against the closed-form vacuum formula.

$\mathcal{C}_+$	$\eta_{\rm crit}$ ($2.8\,M_\odot$, $2\rho_{\rm nuc}$, $\alpha = 0.15$)	$\eta_{\rm crit}$ ($3.2\,M_\odot$, $4\rho_{\rm nuc}$, $\alpha = 0.5$)	$\eta_{\rm vac}$ (Formula I.11)
0.750	0.550	0.547	0.550
0.785	0.663	0.664	0.671
0.830	0.879	0.881	0.902
0.848	1.003	1.003	1.034
0.900	1.633	1.612	1.695
0.945	3.220	3.141	3.337

As shown in Table I.1, the presence of a TOV core alters the critical stiffness by at most $\approx 6\%$ relative to the vacuum formula for the two core models of that table. (A scan of 54 core models in Part II gives -3.8% to $+9.7\%$ for $0.785 \le \mathcal{C}_+ \le 0.845$, and larger deviations at lower $\mathcal{C}_+$ for strongly anisotropic, low-density cores.) Thus, in the stable window, stability is primarily dictated by exterior geometry ($\mathcal{C}_+$).

I.5 Shell Thickness and Self-Consistency

The thin-shell approximation requires $l \ll a$. Estimating physical thickness as $l = \sigma/\rho_s$, where ρ_s is the mean rest-mass density of the shell, we compute l/a across configurations.

Table I.2. Shell thickness estimates for constant shell density ρ_s.

$M\,[M_\odot]$	$\mathcal{C}_+$	a [km]	$\rho_c\,[\rho_{\rm nuc}], \alpha$	$\sigma\,[\text{kg/m}^2]$	l at $\rho_{\rm nuc}$ [km]	l/a	$\rho_s/\rho_{\rm nuc}$ for $l/a = 0.1 \,\vert\, 0.01$	
2.8	0.785	10.53	2; 0.15	3.8×10^{21}	13.9	1.3	13	132
2.8	0.848	9.75	2; 0.15	5.0×10^{21}	18.7	1.9	19	192
2.8	0.848	9.75	4; 0.50	2.9×10^{21}	10.7	1.1	11	110
3.2	0.848	11.14	2; 0.15	4.1×10^{21}	15.3	1.4	14	137
2.8	0.945	8.75	2; 0.15	7.8×10^{21}	28.9	3.3	33	330

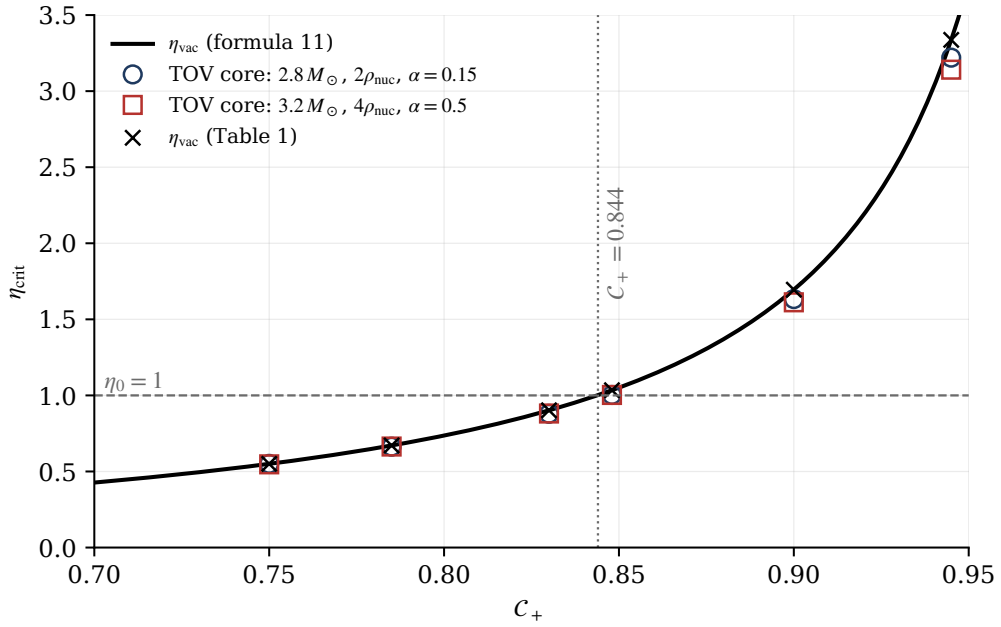


Figure I.1. Critical shell stiffness $\eta_{\rm crit}$ versus $\mathcal{C}_+$ (data of Table I.1) compared with the closed-form vacuum formula.

At nuclear saturation density, a shell carrying 1–2 $M_\odot$ possesses a physical thickness of 10–30 km, yielding ratio $l/a \approx 1.1$–3.3, violating the thin-shell assumption. Self-consistency ($l/a \leq 0.1$) requires extreme shell densities $\rho_s \approx (11\text{–}33)\,\rho_{\rm nuc}$, while $l/a \leq 0.01$ demands $\rho_s \approx (110\text{–}330)\,\rho_{\rm nuc}$. This indicates a sharp density jump at $r = a$ ($\rho_{\rm core} \sim 1\text{–}4\,\rho_{\rm nuc}$ vs $\rho_{\rm shell} \gg \rho_{\rm nuc}$).

Rows with $\mathcal{C}_+ \geq 0.848$ lie above the causal limit $\mathcal{C}_+ \leq 0.844$ of Sec. I.4.1 (the stable, causal window is derived there and in Part II) and are listed to show the trend with compactness; inside the stable, causal window the table contains the row $\mathcal{C}_+ = 0.785$, with $l/a = 1.3$, so the conclusion $l/a \gtrsim 1$ at nuclear density holds there as well.

I.6 Discussion, Scope, and Open Questions

I.6.1 Summary of Findings

1. **Flux Sign:** Rigorously derived from energy-momentum conservation and junction orientation.
2. **Vacuum Limit:** The stability threshold is predominantly determined by exterior compactness $\mathcal{C}_+$, governed by closed-form relation $\eta_{\rm vac}(\mathcal{C}_+)$.
3. **Density Constraint:** Thin-shell self-consistency requires ultra-high densities $\rho_s \geq 11\text{–}33\,\rho_{\rm nuc}$ (for $l/a \leq 0.1$).

I.6.2 Observational Status

It is useful to separate what is established from what is hypothesis. The first steps of the sequence described in the Introduction are standard astrophysics: electron-degenerate matter in white dwarfs, electron capture (the reverse of beta decay) at higher densities, and neutron

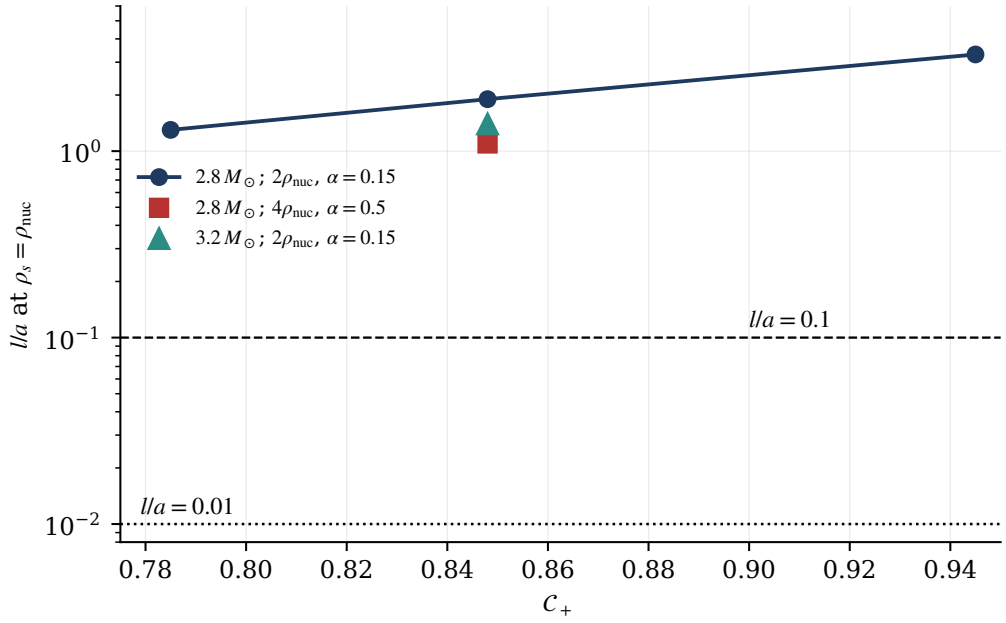


Figure I.2. Shell thickness ratio l/a at $\rho_s = \rho_{\rm nuc}$ versus $\mathcal{C}_+$ (data of Table I.2).

stars. Observations of the most massive pulsars, of neutron-star radii and of tidal deformability from gravitational-wave events constrain the equation of state of dense matter, but they do not determine the state of matter in the innermost core. Deconfinement of quarks beyond the neutron-star stage is therefore an open question, neither confirmed nor excluded by current data, and the final stage of the hypothesis (a mass-generating mechanism switched off) lies far beyond any present observational test.

No compact object with a shell at $\mathcal{C}_+ \approx 0.8$–$0.95$ has been identified. The present paper does not derive observable predictions. Possible tests, to be developed in future work, are signatures of a surface instead of an event horizon, such as gravitational-wave echoes [12], and deviations in the response of such objects to rotation and tides.

I.6.3 Limitations and Future Directions

Collapse Dynamics: The present model establishes static equilibrium and radial stability, but does not describe the dynamical collapse trajectory or mechanism of shell formation.
Static Core Approximation: Full dynamic stability requires coupled perturbation analysis of interior radial modes.
Microphysical Equation of State: Establishing realistic equations of state $P(\sigma)$ for ultra-dense shell matter ($\rho_s \gg \rho_{\rm nuc}$) remains an open challenge.
Rotation and Gravitational Waves: Future extensions will incorporate slow rotation up to $\mathcal{O}(\Omega^2)$ via the Hartle–Thorne formalism [8, 13] and evaluate gravitational wave echo signatures [12].

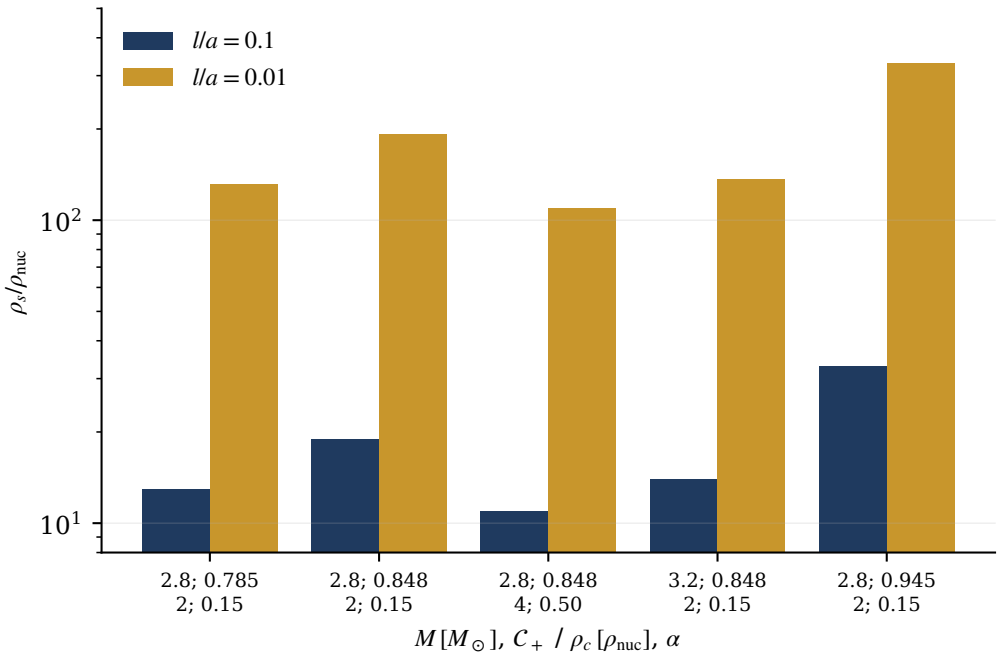


Figure I.3. Shell density $\rho_s/\rho_{\rm nuc}$ required for $l/a = 0.1$ and $l/a = 0.01$ (data of Table I.2).

PART II

Equation of State of the Anisotropic Thin Shell: Exact Stability Window, No-Go for Linear Closures, and an Exterior Echo-Delay Estimate

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Part II of "Anisotropic Thin Shell Around a 3 $M_\odot$ Core" — November 5, 2026

Abstract. In Part I we showed that a static anisotropic thin shell around a regular core can exist for exterior compactness $\mathcal{C}_+ = 2GM/(c^2 a) \in [0.785, 0.945]$, and that radial stability requires a shell stiffness $\eta_0 = dP/d\sigma$ larger than a critical value $\eta_{\rm vac}(\mathcal{C}_+)$. Part I left open which surface equation of state $P(\sigma)$ can satisfy this. Here we answer that question within the model. We (i) re-derive the closed form of $\eta_{\rm vac}$ and obtain the exact second variation of the effective potential, $V''(a_0) = 2(1+3y)(\eta_0 - \eta_{\rm vac})/a_0^2$ with $y = \sqrt{1-\mathcal{C}_+}$; (ii) prove that every linear closure $P = w\sigma$ is unstable at every compactness, because the equilibrium value of $P_0/\sigma_0 = (1-y)/(4y)$ is always smaller than $\eta_{\rm vac}$; (iii) show that, in the vacuum-interior limit, a polytropic surface law $P \propto \sigma^\Gamma$ is stable and causal only if Γ lies in the window $\Gamma_{\min}(y) < \Gamma \le \Gamma_{\max}(y)$ with $\Gamma_{\min} = (1+2y+3y^2)/[y(1+3y)]$ and $\Gamma_{\max} = 4y/(1-y)$, and that this window closes exactly at the root of the cubic $15y^3 + 3y^2 - y - 1 = 0$, i.e. $\mathcal{C}_+^* = 0.8437$, which is the origin of the number 0.844 of Part I (with the TOV cores of Part I the equilibrium ratio P_0/σ_0 is lower, so the window moves to larger Γ, e.g. 3.7–5.5 at $\mathcal{C}_+ = 0.785$ for the 2.8 $M_\odot$ core, and closes at $\mathcal{C}_+ \simeq 0.8476$); (iv) give an exterior-geometry lower bound on the echo delay, $\Delta t \simeq 49\,\mu{\rm s}\,(M/3\,M_\odot)$ at $\mathcal{C}_+ = 0.785$. The analytic results are confirmed by an independent numerical integration of the shell equations, and Table I.1 of Part I, including its flux term, was independently reproduced. The microphysical origin of such a stiff shell material remains open.

II.1 Purpose and relation to Part I

Part I [1] studied a model in which collapsing matter of mass $M \sim 3\,M_\odot$ is stopped by a thin shell with anisotropic stresses surrounding a regular core. Its main outputs were: the DEC-admissible compactness window, the closed-form critical stiffness $\eta_{\rm vac}(\mathcal{C}_+)$ (its Eq. (I.11)), and the thickness constraint $\rho_s \gtrsim 11$–$33\,\rho_{\rm nuc}$. The limitation list of Part I names, among others, the absence of a surface equation of state $P(\sigma)$ for the shell material. This note addresses that point in the only way that is possible without a microphysical model: it asks which classes of $P(\sigma)$ are compatible with stability, causality and the dominant energy condition, and what this implies for the shell matter.

Figure II.1 shows the geometry, using the units $G = c = 1$ throughout.

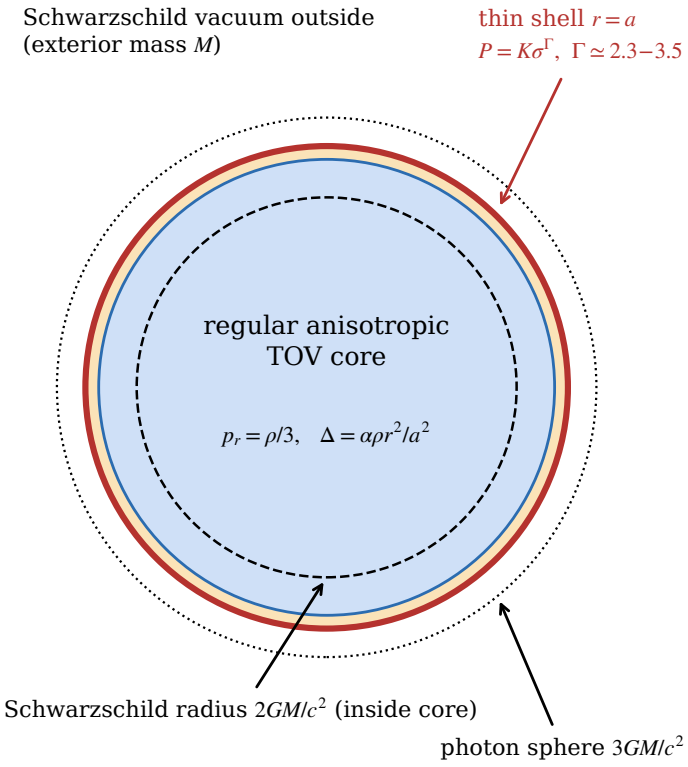


Figure II.1. Geometry of the model at $\mathcal{C}_+ = 0.785$ (shell thickness exaggerated). The shell sits inside the photon sphere and outside the Schwarzschild radius, which lies in the interior of the regular core.

II.2 Shell dynamics in the vacuum-interior limit

Let $a(\tau)$ be the shell radius, σ and P its surface energy density and pressure, M the exterior mass, and define

$$s = 4\pi a\sigma, \qquad \Pi = 8\pi a P, \qquad \mathcal{C}_+ = \frac{2M}{a}, \qquad y = \sqrt{1-\mathcal{C}_+}. \tag{II.1}$$

For a vacuum interior ($m_- = 0$, as in Part I, Sec. I.4.1) the Darmois–Israel junction conditions give the equation of motion $\dot a^2 + V(a) = 0$ with

$$V(a) = 1 - A^2, \qquad A(a) = \frac{1}{2}\left[s + \frac{2M}{a\,s}\right]. \tag{II.2}$$

Energy conservation on the shell (no flux in the vacuum limit), $(\sigma a^2)' = -P\,(a^2)'$, gives

$$s' = -\frac{s+\Pi}{a}, \qquad \Pi' = \frac{\Pi}{a} + 2\eta\left(s' - \frac{s}{a}\right), \qquad \eta \equiv \frac{dP}{d\sigma}, \tag{II.3}$$

where the second relation follows from $\Pi = 8\pi a P$ and $\sigma = s/(4\pi a)$.

Equilibrium. The conditions $V = 0$ and $V' = 0$ are equivalent to $A = 1$ and $A' = 0$. The first

gives $s^2 - 2s + \mathcal{C}_+ = 0$, hence

$$s_0 = 1 - y, \qquad \Pi_0 = \frac{(1-y)^2}{2y}, \qquad \text{so} \qquad w_0 \equiv \frac{P_0}{\sigma_0} = \frac{\Pi_0}{2s_0} = \frac{1-y}{4y}. \tag{II.4}$$

(The value of Π_0 follows from $A' = \frac{1}{2}\left[s'(1 - \mathcal{C}_+/s^2) - \mathcal{C}_+/(as)\right] = 0$ with s' from (II.3) and $\mathcal{C}_+ = 2s_0 - s_0^2$.) Note that w_0 is a *prediction* of the model: it is the ratio of tangential pressure to energy density that the shell matter must have in equilibrium at a given compactness.

Exact second variation. At equilibrium $V'' = -2A''$. Differentiating A twice and inserting (II.3), (II.4) (details in Appendix A) gives

$$V''(a_0) = \frac{2(1+3y)}{a_0^2}\,(\eta_0 - \eta_{\rm vac})\,, \qquad \eta_{\rm vac}(y) = \frac{(1-y)(1+2y+3y^2)}{4y^2(1+3y)}. \tag{II.5}$$

Equation (II.5) reproduces Eq. (I.11) of Part I, and adds two facts that were not stated there: the threshold is exact (not only a root of $A'' = 0$ but a factor of V''), and the prefactor $2(1+3y)/a_0^2$ is strictly positive, so $\eta_0 > \eta_{\rm vac}$ is both necessary and sufficient for $V'' > 0$ in the vacuum limit.

Interpretation of η_0. If the shell is thin and its thickness l is held fixed during a small perturbation, then $\sigma = \rho_s l$ and $P = \langle p_t \rangle l$, so that $\eta_0 = dp_t/d\rho_s$ is an effective squared tangential sound speed, and $w_0 = \langle p_t \rangle / \rho_s$. The causality requirement is $\eta_0 \leq 1$, the dominant energy condition requires $w_0 \leq 1$. This reading is an assumption of the thin-shell picture, not a derived result.

II.3 Results

II.3.1 Critical stiffness and special compactnesses

Figure II.2 and Table II.1 summarise $\eta_{\rm vac}(\mathcal{C}_+)$ and $w_0(\mathcal{C}_+)$. Three special values of the compactness follow from simple algebra:

- $\eta_{\rm vac} = 1/3$ at the root of $21y^3 + y^2 - 3y - 3 = 0$, i.e. $\mathcal{C}_+ \simeq 0.646$. Above this compactness the shell matter must be stiffer than conformal matter ($c_s^2 > 1/3$).
- $w_0 = 1/3$ at $y = 3/7$, i.e. exactly $\mathcal{C}_+ = 40/49 \simeq 0.816$. There the shell's ratio of tangential pressure to density equals the value $\rho/3$ used for the radial closure of the core. This is a curiosity noted for the record; the present model does not derive it.
- $\eta_{\rm vac} = 1$ at the root of

$$15y^3 + 3y^2 - y - 1 = 0, \qquad y_* = 0.39529, \qquad \mathcal{C}_+^* = 1 - y_*^2 = 0.84374. \tag{II.6}$$

  Equation (II.6) follows from $(1-y)(1+2y+3y^2) = 4y^2(1+3y)$. Thus the number $\mathcal{C}_+ \leq 0.844$ of Part I is not a numerical accident: it is the compactness at which the critical stiffness reaches the speed of light.

II.3.2 No-go for linear closures

Proposition 1 (Linear closure)**.** *In the vacuum limit, no shell with $P = w\sigma$ (constant w) is radially stable at any compactness $\mathcal{C}_+ \in (0,1)$.*

Table II.1. Vacuum-limit stability data. $\Gamma_{\min}$ and $\Gamma_{\max}$ are defined in Sec. II.3.3.

$\mathcal{C}_+$	y	$w_0 = P_0/\sigma_0$	$\eta_{\rm vac}$	$\Gamma_{\min}$	$\Gamma_{\max}$
0.600	0.6325	0.1453	0.2747	1.891	6.883
0.646	0.5950	0.1702	0.3340	1.963	5.876
0.700	0.5477	0.2064	0.4271	2.069	4.844
0.750	0.5000	0.2500	0.5500	2.200	4.000
0.785	0.4637	0.2892	0.6709	2.320	3.458
0.8163	0.4286	0.3333	0.8193	2.458	3.000
0.830	0.4123	0.3563	0.9020	2.531	2.806
0.8437	0.3953	0.3824	1.0000	2.615	2.615

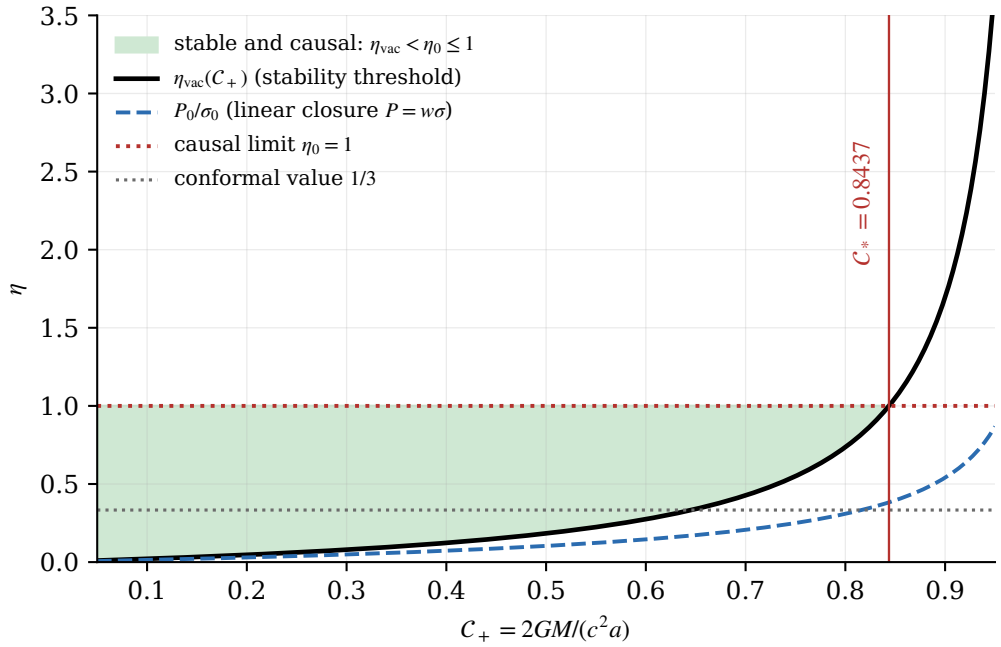


Figure II.2. Critical stiffness $\eta_{\rm vac}$ (black), equilibrium ratio P_0/σ_0 (dashed), causal limit 1, and the stable-and-causal region (green). Note that the dashed curve lies below the black one everywhere.

Proof. For $P = w\sigma$ one has $\eta_0 = w$, and equilibrium forces $w = w_0 = (1-y)/(4y)$ by (II.4). Then

$$\frac{\eta_{\rm vac}}{w_0} = \frac{1+2y+3y^2}{y(1+3y)} = 1 + \frac{1+y}{y(1+3y)} > 1 \quad \text{for } y \in (0,1), \tag{II.7}$$

so $\eta_0 = w_0 < \eta_{\rm vac}$ and $V'' < 0$ by (II.5). □

The stability of a hypothetical shell therefore cannot be obtained from a "constant equation-of-state parameter" closure, which is the first thing one would try. The shell matter must have a pressure that grows faster than its density.

II.3.3 Polytropic window

Take $P = K\sigma^{\Gamma}$, so that $\eta_0 = \Gamma\, P_0/\sigma_0 = \Gamma w_0$. Stability $\eta_0 > \eta_{\rm vac}$ and causality $\eta_0 \le 1$ translate into

$$\Gamma_{\min}(y) = \frac{\eta_{\rm vac}}{w_0} = \frac{1+2y+3y^2}{y(1+3y)} < \Gamma \le \Gamma_{\max}(y) = \frac{1}{w_0} = \frac{4y}{1-y}. \tag{II.8}$$

Proposition 2 (Properties of the window, vacuum interior). *1.* $\Gamma_{\min} > 3/2$ *for all* $\mathcal{C}_+ \in (0,1)$*; the bound* $3/2$ *is approached as* $\mathcal{C}_+ \to 0$*. Indeed* $\Gamma_{\min} - \frac{3}{2} = \dfrac{(1-y)(2+3y)}{2y(1+3y)} > 0$.

2. *The window is non-empty if and only if* $\mathcal{C}_+ \le \mathcal{C}_+^* = 0.84374$*, the root of* (II.6)*. At* $\mathcal{C}_+^*$ *it closes to the single value* $\Gamma \simeq 2.615$.

3. *Over the DEC window of Part I that is also causal* ($0.785 \le \mathcal{C}_+ \le \mathcal{C}_+^*$)*,* Γ *must lie between* 2.32 *and* 3.46 *at the lower end and become tightly constrained* ($\Gamma \in [2.53, 2.81]$ *at* $\mathcal{C}_+ = 0.83$) *towards the upper end.*

Proof. (1) is the displayed identity. (2) $\Gamma_{\min} \le \Gamma_{\max}$ is equivalent to $\eta_{\rm vac} \le 1$, i.e. to $(1-y)(1+2y+3y^2) \le 4y^2(1+3y)$, i.e. to $15y^3 + 3y^2 - y - 1 \ge 0$ (note the sign: the cubic is increasing in y on $(0,1)$ and vanishes at y_*, so the condition holds for $y \ge y_*$, i.e. $\mathcal{C}_+ \le \mathcal{C}_+^*$). (3) is read off from Table II.1. □

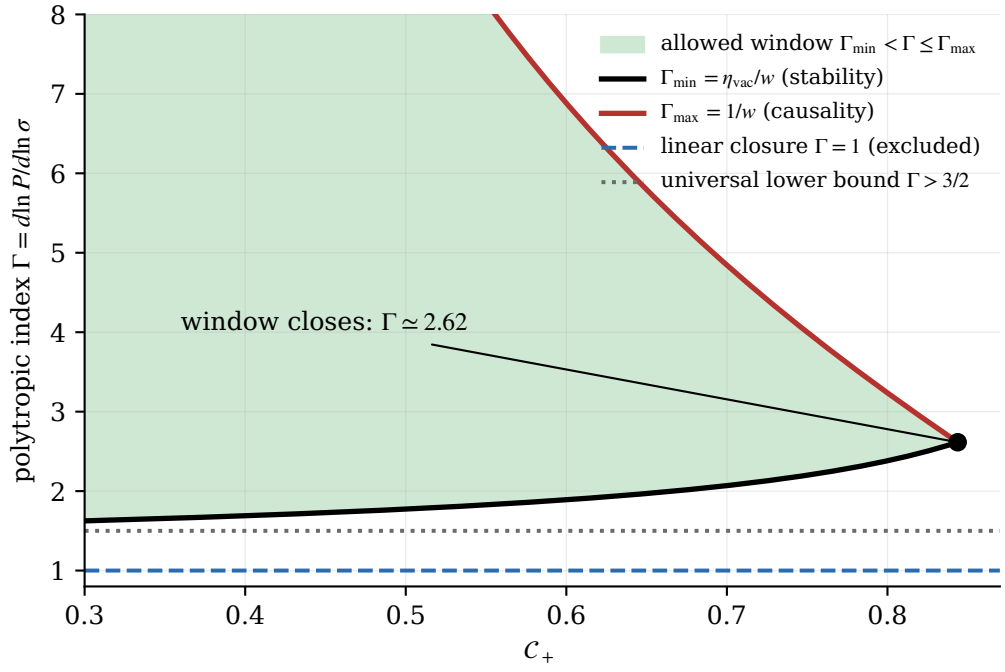


Figure II.3. Allowed polytropic index Γ of the shell: green region between the stability bound (black) and the causality bound (red). The window narrows to a point at $\mathcal{C}_+^*$. The linear closure $\Gamma = 1$ is excluded everywhere.

Physical reading. In the stable region the effective squared sound speed of the shell material is $c_s^2 \equiv \eta_0 \in (0.55, 1]$ at $\mathcal{C}_+ \ge 0.75$, compared with $1/3$ for conformal (radiation-like) matter and ≈ 0.2–0.5 for typical neutron-star equations of state. The model therefore requires an extremely stiff, nearly causal shell material, in addition to the very high density $\rho_s \gtrsim 11$–$33\,\rho_{\rm nuc}$ already

required by thin-shell self-consistency (Part I, Sec. I.5, for $l/a \leq 0.1$). Both requirements point the same way: a conventional hadronic equation of state is not what this shell is made of, and the hypothesis stands or falls on whether a physically motivated ultra-dense phase can supply $\Gamma \approx 2.3$–3.5 (vacuum-interior limit; with the cores of Part I the required Γ is larger, Sec. II.3.5).

II.3.4 Independent numerical check

To guard against algebra errors, the system (II.3) with $P = K\sigma^\Gamma$ was integrated numerically (adaptive Runge–Kutta, relative tolerance 10^{-13}) from equilibrium data (II.4), $A(a)$ was formed along the solution, and V'' was obtained by a second-order finite difference ($\delta a = 10^{-3} a_0$). Table II.2 compares the result with (II.5) for $\Gamma = \Gamma_{\min} \pm 0.15$ and for $\Gamma = 1$. The sign of V'' flips at $\Gamma_{\min}$ (the value at $\Gamma = \Gamma_{\min}$ itself is $\lesssim 10^{-5}$, i.e. zero within discretisation error), and the numbers agree with the closed form to three or four digits.

Table II.2. Numerical integration of the shell equations against the analytic second variation. A dash means the closed-form value was not tabulated for that entry.

$\mathcal{C}_+$	$\Gamma_{\min}$	$a_0^2 V''$ at $\Gamma_{\min} - 0.15$ numeric / Eq. (II.5)	$a_0^2 V''$ at $\Gamma_{\min} + 0.15$ numeric / Eq. (II.5)	$a_0^2 V''$ at $\Gamma = 1$ numeric / Eq. (II.5)
0.700	2.069	-0.1637 / -0.1637	$+0.1637$ / $+0.1637$	-1.1667 / -1.1666
0.785	2.320	-0.2074 / -0.2075	$+0.2074$ / $+0.2075$	-1.8256 / -1.8254
0.8163	2.458	-0.2286 / -0.2286	$+0.2285$ / $+0.2286$	-2.2218 / –
0.830	2.531	-0.2391 / -0.2391	$+0.2391$ / $+0.2391$	-2.4412 / –
0.840	2.591	-0.2475 / -0.2475	$+0.2475$ / $+0.2475$	-2.6250 / –

II.3.5 Effect of a TOV core, and independent check of Part I

The shell equations of Part I with a regular anisotropic core (Eqs. (I.5), (I.6) and (I.10) of Part I, static-core assumption) were re-implemented independently (DOP853 for the core, finite-difference V'' along the shell trajectory). The result reproduces Table I.1 of Part I for both core models in all twelve entries to within 5×10^{-4}, for example $\eta_{\rm crit} = 0.6626$ ($2.8\,M_\odot$, $2\rho_{\rm nuc}$, $\alpha = 0.15$, $\mathcal{C}_+ = 0.785$; Part I: 0.663) and 3.1416 ($3.2\,M_\odot$, $4\rho_{\rm nuc}$, $\alpha = 0.5$, $\mathcal{C}_+ = 0.945$; Part I: 3.141).

The same code also checks the flux term. At equilibrium the first derivative $V'(a_0)$ must vanish when σ_0 and P_0 are taken from the independent junction conditions (I.5), (I.6). With the flux sign of Part I, Eq. (I.10), one finds $a_0 V' = 6 \times 10^{-8}$ (zero within discretisation error); with the opposite sign $a_0 V' = 2.7$ and $\eta_{\rm crit} = 14.8$. The sign of the flux term derived in Part I is therefore confirmed by an independent consistency condition.

For the two core models of Part I the core changes the critical stiffness by -0.6% to -5.9% over $0.75 \leq \mathcal{C}_+ \leq 0.945$, consistent with the $\approx 6\%$ stated in Part I for these two models. Over a wider family of 54 core models ($M = 2.8$–$3.2\,M_\odot$, $\rho_c = 1$–$4\,\rho_{\rm nuc}$, $\alpha = 0$–0.5) the deviation from $\eta_{\rm vac}$ lies between -3.8% and $+9.7\%$ for $0.785 \leq \mathcal{C}_+ \leq 0.845$, but reaches $+18\%$ at $\mathcal{C}_+ = 0.75$ and $+33\%$ at $\mathcal{C}_+ = 0.70$ for the low-density, strongly anisotropic core ($2.8\,M_\odot$, $\rho_c = \rho_{\rm nuc}$, $\alpha = 0.5$). The core changes the equilibrium ratio $w_0 = P_0/\sigma_0$ much more strongly, and with it the polytropic window of Sec. II.3.3.

Table II.3 shows the first core model of Part I. At $\mathcal{C}_+ = 0.785$ the ratio is $w_0 = 0.18$ instead of 0.29, so Γ must lie in $[3.65, 5.51]$ instead of $[2.32, 3.46]$. The window closes where $\eta_{\rm crit} = 1$, at $\mathcal{C}_+ \simeq 0.8476$ (vacuum: 0.8437); the second core model closes at 0.8475. Over 12 corner models of

the family the closing point lies between 0.8442 and 0.8490. For the second model ($3.2\,M_\odot$, $4\rho_{\rm nuc}$, $\alpha = 0.5$) the equilibrium pressure is negative (tension) for $\mathcal{C}_+ \lesssim 0.83$ ($w_0 = -0.79, -0.40, -0.04$ at $\mathcal{C}_+ = 0.75, 0.785, 0.83$), so a polytropic law $P \propto \sigma^\Gamma$ with $P_0 < 0$ is not meaningful there and a linear form $P = P_0 + \eta_0(\sigma - \sigma_0)$ must be used.

Table II.3. Core model $2.8\,M_\odot$, $\rho_c = 2\rho_{\rm nuc}$, $\alpha = 0.15$: equilibrium ratio and Γ window with the TOV core, compared with the vacuum-interior values. The stability condition $\eta_0 > \eta_{\rm crit}$ is almost unchanged, but the required Γ is not.

$\mathcal{C}_+$	w_0 (core)	w_0 (vacuum)	$\eta_{\rm crit}$ (core)	$\Gamma_{\rm min}$	$\Gamma_{\rm max}$
0.750	0.101	0.250	0.5504	5.45	9.90
0.785	0.182	0.289	0.6626	3.65	5.51
0.830	0.293	0.356	0.8787	3.00	3.41
0.848	0.344	0.391	1.0032	window closed	

Consequently, the statements of this paper that concern η_0 (the threshold $\eta_{\rm vac}$, the no-go for constant w, the need for c_s^2 well above $1/3$ at $\mathcal{C}_+ \gtrsim 0.7$) change by $\sim \pm 10\%$ or less in the stable window $0.785 \le \mathcal{C}_+ \le 0.845$ when a core is added, whereas those that concern Γ and w_0 (Secs. II.3.2–II.3.3) are results for the vacuum-interior limit and shift when a core is present.

II.4 Exterior echo-delay estimate

If the object has a reflecting surface at $r = a$ instead of a horizon, gravitational-wave echoes [12] return after approximately twice the light-travel time between the surface and the effective potential barrier near the photon sphere. Using the Schwarzschild tortoise coordinate $r_* = r + 2M\ln(r/2M - 1)$ and approximating the barrier peak by $r = 3M$,

$$\Delta t_{\rm echo} \simeq 2\left[r_*(3M) - r_*(a)\right], \qquad a = \frac{2M}{\mathcal{C}_+}. \tag{II.9}$$

Table II.4 gives values with $GM_\odot/c^3 = 4.9255\,\mu$s. The estimate is a *lower bound* on the full delay, because it neglects travel time inside the shell and the core, and it is linear in M.

Table II.4. Exterior-geometry echo delay (lower bound).

$\mathcal{C}_+$	$a/(GM/c^2)$	$\Delta t/(GM/c^3)$	Δt for $3\,M_\odot$ [μs]
0.750	2.667	2.29	33.8
0.785	2.548	3.31	48.9
0.8163	2.450	4.29	63.4
0.8437	2.371	5.23	77.3
0.900	2.222	7.57	111.9
0.945	2.116	10.37	153.2

For $M = 3\,M_\odot$ this is tens of microseconds, i.e. a characteristic frequency of order 10^4 Hz, above the sensitive band of current ground-based detectors; for heavier remnants of the same structure the delay scales linearly with the mass (for example ≈ 0.5 ms at $30\,M_\odot$ and $\mathcal{C}_+ = 0.785$). The estimate is therefore a definite, falsifiable scaling statement rather than an immediate observational claim.

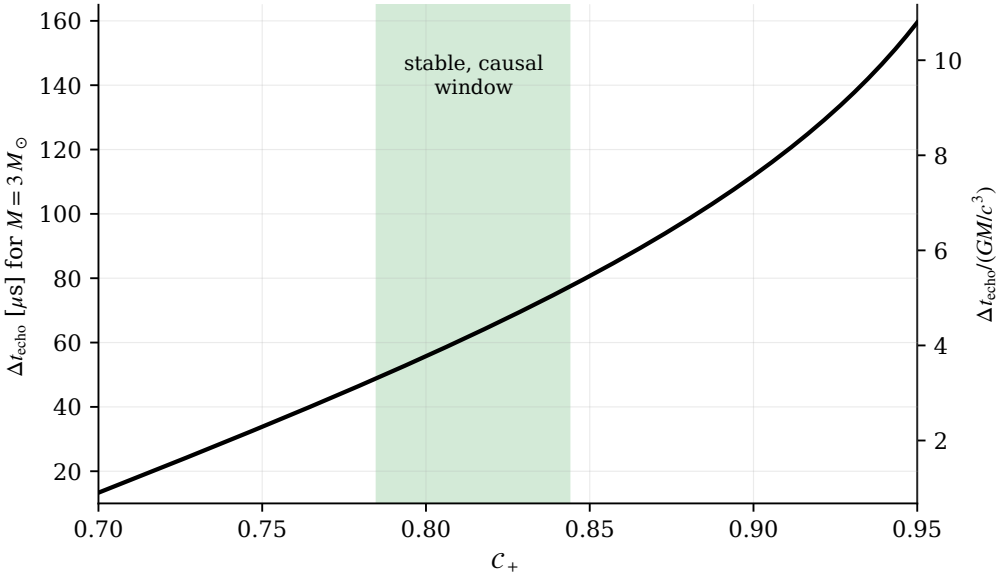


Figure II.4. Echo delay versus compactness for $M = 3\,M_\odot$. The shaded band is the stable, causal window.

II.5 Discussion

What has been established. Within the vacuum-interior limit of the thin-shell model: (a) the stability threshold is exactly $\eta_0 = \eta_{\rm vac}$; (b) constant-w closures are excluded; (c) in the vacuum-interior limit polytropic shells need Γ in an explicit window that closes at $\mathcal{C}_+^* = 0.8437$ (with the cores of Part I the window moves to larger Γ and closes at $\simeq 0.8476$); (d) the shell material must be stiffer than conformal matter above $\mathcal{C}_+ \simeq 0.646$ and nearly causal near the top of the window.

What has not been established. The model does not explain why matter at $\rho_s \gg \rho_{\rm nuc}$ should have $\Gamma \approx 2.3$–3.5; it contains no collapse dynamics; the core is treated as static; the thin-shell approximation is itself marginal at $\rho_s \sim \rho_{\rm nuc}$ (Part I, Table I.2). The identification of η_0 with a tangential sound speed relies on the fixed-thickness assumption. The link to the Standard-Model mass-generation scales of Part I (Introduction) is not tested here.

Sharpest next steps. (1) Replace the thin shell by a finite-thickness layer with the same (Γ, w_0) and test whether $\Gamma \approx 2.3$–3.5 is compatible with the integrated TOV structure with anisotropy. (2) Repeat the stability analysis with the full core perturbations, instead of the static-core assumption. (3) Compare the required Γ with equations of state of deconfined or colour-superconducting quark matter, where stiffness close to the causal limit is discussed in the literature. (4) Compute the slow-rotation response (Hartle–Thorne) of the stable configurations.

Acknowledgments and reproducibility

The symbolic derivation of Eq. (II.5), the closed forms of Sec. II.3, the numerical integration of Table II.2, the independent reproduction of Part I, Table I.1 and the echo estimate were carried out with the assistance of an AI system (Claude, Anthropic) under the direction of the author, who is responsible for the content. Every closed-form statement above was verified

symbolically (computer algebra) and, for the stability claims, numerically by an independent method. The physical hypothesis and the modelling choices are the author's.

A Derivation of the second variation

Write $\mathcal{C}_+(a) = 2M/a$. Differentiating (II.2),

$$A' = \frac{1}{2}\left[s'\left(1 - \frac{\mathcal{C}_+}{s^2}\right) - \frac{\mathcal{C}_+}{a\,s}\right]. \tag{II.10}$$

Differentiating once more, replacing s' and Π' by (II.3) and evaluating at the equilibrium (II.4) (where $A = 1$, $A' = 0$, $s_0 = 1 - y$, $\mathcal{C}_+ = 1 - y^2$), a symbolic computation gives

$$A''(a_0) = -\frac{12\eta y^3 + 4\eta y^2 + 3y^3 - y^2 - y - 1}{4a_0^2 y^2} = -\frac{4y^2(1+3y)\,\eta - (1-y)(1+2y+3y^2)}{4a_0^2 y^2}. \tag{II.11}$$

Since $V = 1 - A^2$, $V'' = -2(A'^2 + AA'') = -2A''$ at equilibrium, which is Eq. (II.5). Setting $A'' = 0$ reproduces the closed form of Part I; the linear dependence on η shows that this root is the only threshold.

B Short list of the formulae used

$$y = \sqrt{1 - \mathcal{C}_+}, \quad s_0 = 1 - y, \quad \frac{P_0}{\sigma_0} = \frac{1-y}{4y}, \quad \eta_{\text{vac}} = \frac{(1-y)(1+2y+3y^2)}{4y^2(1+3y)},$$

$$\Gamma_{\min} = \frac{1+2y+3y^2}{y(1+3y)}, \quad \Gamma_{\max} = \frac{4y}{1-y}, \quad 15y_*^3 + 3y_*^2 - y_* - 1 = 0 \;\Rightarrow\; \mathcal{C}_+^* = 0.84374.$$

PART III

A Constant-Sound-Speed Realisation of the Stable Anisotropic Thin Shell and the Required Vacuum-Offset Energy

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Part III of "Anisotropic Thin Shell Around a 3 $M_\odot$ Core" — November 2026

Abstract. Part II showed that a stable, causal shell requires an effective stiffness $\eta_0 > \eta_{\rm vac}(\mathcal{C}_+)$ and that linear closures $P = w\sigma$ are excluded. Here we test the simplest physically motivated family that escapes this no-go: the constant-sound-speed (CSS) law $p_t = c_s^2(\rho - \rho_*)$, in which a fixed fraction of the energy density does not bear pressure. We show that within the vacuum-interior model (i) a CSS shell is radially stable if and only if $c_s^2 > \eta_{\rm vac}$, with the equilibrium offset fixed exactly by $\rho_*/\rho_s = 1 - w_0/c_s^2$, $w_0 = (1-y)/(4y)$; (ii) every stable CSS shell has $\rho_*/\rho_s > 1 - 1/\Gamma_{\min} > 1/3$, in practice 0.49–0.62 over $0.646 \le \mathcal{C}_+ \le 0.844$; (iii) at $\mathcal{C}_+ = 0.785$ and the shell density $\rho_s \approx 13\,\rho_{\rm nuc}$ required by Part I for $l/a = 0.1$ this means $\rho_* \approx 7.6$–$9.2\,\rho_{\rm nuc}$ (other estimates of ρ_s at the same $\mathcal{C}_+$ are compared in Sec. III.3). The shell therefore must be made of matter in which more than one third, and typically more than one half, of the energy density is not pressure-bearing, which is the quantitative content of the statement that the shell material cannot be an ordinary compressed fluid. (iv) With the TOV cores of Part I (54 core models scanned, $M = 2.8$–$3.2\,M_\odot$, $\rho_c = 1$–$4\,\rho_{\rm nuc}$, $\alpha = 0$–0.5) the bound is stronger, $f > 0.61$ at $\mathcal{C}_+ = 0.785$ and $f > 0.62$ at $\mathcal{C}_+ = 0.83$, and strongly anisotropic cores have a negative equilibrium shell pressure (tension) and require $\rho_* > \rho_s$. The microscopic origin of such an offset is not derived here.

III.1 Motivation

Part II reduced the shell's stability to one condition on its sound speed, $\eta_0 = dP/d\sigma > \eta_{\rm vac}(\mathcal{C}_+)$, with equilibrium ratio $w_0 = P_0/\sigma_0 = (1-y)/(4y)$, $y = \sqrt{1-\mathcal{C}_+}$. A pure power law $P \propto \sigma^\Gamma$ needs $\Gamma \gtrsim 2.3$, which has no obvious microscopic counterpart. A natural alternative, widely used for dense and deconfined matter, is the CSS parametrisation. Its structure matches the intuition behind the hypothesis of Part I (compression strips matter of the structure built by the interactions): a CSS fluid carries an energy density ρ_* that is released but contributes no pressure, like a bag constant. This note works out what that family implies for the shell.

III.2 CSS shell: exact relations

Under the fixed-thickness reading of Part II, $\sigma = \rho_s l$ and $P = p_t l$. Take

$$p_t = c_s^2(\rho - \rho_*), \qquad 0 < c_s^2 \le 1, \qquad 0 \le \rho_* < \rho_s. \tag{III.1}$$

Then $\eta_0 = dp_t/d\rho = c_s^2$ for any state, and the equilibrium ratio is $w_0 = p_t/\rho = c_s^2(1-\rho_*/\rho_s)$, i.e.

$$f \equiv \frac{\rho_*}{\rho_s} = 1 - \frac{w_0}{c_s^2}. \tag{III.2}$$

Proposition 3. *In the vacuum-interior limit, a CSS shell at compactness $\mathcal{C}_+$ is radially stable and causal if and only if $\eta_{\rm vac}(\mathcal{C}_+) < c_s^2 \le 1$ (so $\mathcal{C}_+ \le \mathcal{C}_+^* = 0.8437$), and its offset fraction is then given by Eq. (III.2), with*

$$1 - \frac{1}{\Gamma_{\min}} < f \le 1 - w_0, \qquad \Gamma_{\min} = \frac{1+2y+3y^2}{y(1+3y)}. \tag{III.3}$$

Proof. Stability is $\eta_0 > \eta_{\rm vac}$ (Part II, Eq. (II.5)) with $\eta_0 = c_s^2$, and causality is $c_s^2 \le 1$. By (III.2), f increases with c_s^2; its infimum over $c_s^2 > \eta_{\rm vac}$ is $1 - w_0/\eta_{\rm vac} = 1 - 1/\Gamma_{\min}$ and its maximum, at $c_s^2 = 1$, is $1 - w_0$. □

Corollary 1. *Since $\Gamma_{\min} > 3/2$ for every $\mathcal{C}_+$ (Part II), every stable CSS shell has $\rho_* > \rho_s/3$.*

Conformal matter has $c_s^2 = 1/3$ and $\rho_* = 0$, and is excluded for $\mathcal{C}_+ > 0.646$; the corollary says that the offset is not a fine-tuning but a structural requirement.

III.3 Numbers

Table III.1. Allowed range of the offset fraction $f = \rho_*/\rho_s$ for CSS shells. The range collapses to $f = 0.618$ at $\mathcal{C}_+^*$.

$\mathcal{C}_+$	w_0	$\eta_{\rm vac}$	f lower bound $1 - 1/\Gamma_{\min}$	f at $c_s^2 = 1$
0.646	0.1702	0.3340	0.491	0.830
0.700	0.2064	0.4271	0.517	0.794
0.750	0.2500	0.5500	0.546	0.750
0.785	0.2892	0.6709	0.569	0.711
0.8163	0.3333	0.8193	0.593	0.667
0.830	0.3563	0.9020	0.605	0.644
0.8437	0.3824	1.0000	0.618	0.618

Absolute scale at $\mathcal{C}_+ = 0.785$. For $M = 2.8\,M_\odot$ ($a = 10.53$ km) Part I requires $\rho_s \approx 13\,\rho_{\rm nuc}$ for $l/a = 0.1$ (a vacuum-limit estimate without core gives $19\,\rho_{\rm nuc}$). With $\rho_{\rm nuc}c^2 \simeq 151$ MeV/fm^3:

c_s^2	$f = \rho_*/\rho_s$	$\rho_*/\rho_{\rm nuc}$	$\rho_* c^2$ [MeV/fm^3]
0.70	0.587	7.6	1160
0.80	0.639	8.3	1260
0.90	0.679	8.8	1340
1.00	0.711	9.2	1400

For orientation: in the MIT-bag description of quark matter the offset is $\rho_* = 4B$, and the commonly used range $B^{1/4} \approx 145$–200 MeV gives $4B \approx 230$–830 MeV/fm^3, i.e. ~ 1.5–$5.5\,\rho_{\rm nuc}$. The offset scales linearly with the adopted ρ_s. The same configuration has three estimates of ρ_s for $l/a = 0.1$: $13\,\rho_{\rm nuc}$ (Part I, with core), $15.5\,\rho_{\rm nuc}$ (Part IV, finite uniform layer with vacuum interior) and $19\,\rho_{\rm nuc}$ (vacuum-limit thin-shell estimate). With $f = 0.587$–0.711 they

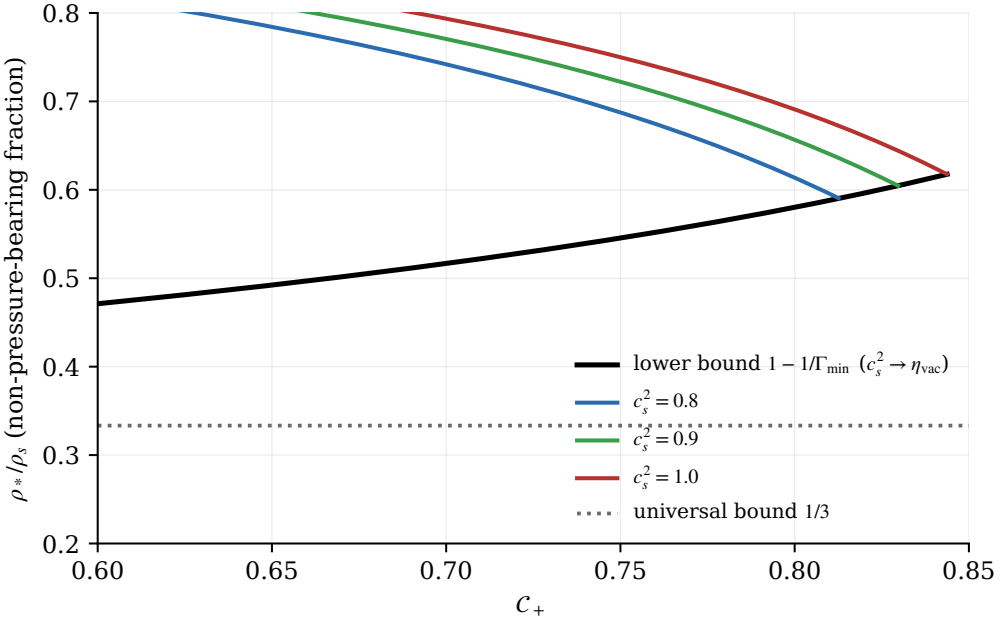


Figure III.1. Offset fraction ρ_*/ρ_s versus compactness: black is the stability lower bound, coloured curves are fixed c_s^2 (drawn only where stable). The window closes at $\mathcal{C}_+^*$.

give $\rho_* \approx 7.6$–9.2, 9.1–11.0 and 11.2–$13.5\,\rho_{\rm nuc}$ respectively; the qualitative conclusion ($f > 1/3$, typically $f > 1/2$) does not depend on this choice. The comparison with the bag constant is an order-of-magnitude comparison made from memory of standard values and should be checked against the literature before it is quoted. If it holds, the shell requires an offset several times larger than a standard bag constant at the same $\mathcal{C}_+$, unless the shell is thicker than the thin-shell limit (smaller ρ_s).

III.4 Effect of the TOV core

Sections III.2–III.3 assume a vacuum interior. For the cores of Part I the equilibrium ratio $w_0 = P_0/\sigma_0$ and the critical stiffness were recomputed (junction conditions (I.5), (I.6) and flux equation (I.10) of Part I, static core; the code reproduces Table I.1 of Part I to 5×10^{-4}, see Part II). The stability threshold is only moderately sensitive to the core: $\eta_{\rm crit} = 0.653$–0.736 at $\mathcal{C}_+ = 0.785$ and 0.876–0.917 at $\mathcal{C}_+ = 0.83$ over all 54 models, within -2.7% to $+9.7\%$ ($\mathcal{C}_+ = 0.785$, $\eta_{\rm vac} = 0.671$) and -2.9% to $+1.6\%$ ($\mathcal{C}_+ = 0.83$, $\eta_{\rm vac} = 0.902$) of the vacuum value. The equilibrium ratio is not: w_0 ranges from -0.40 to $+0.26$ at $\mathcal{C}_+ = 0.785$, decreasing with the core anisotropy α. Because $f_{\min} = 1 - w_0/\eta_{\rm crit}$, the offset requirement grows accordingly.

Over the scanned family, every model with a positive equilibrium pressure needs $f > 0.61$ ($\mathcal{C}_+ = 0.785$) or $f > 0.62$ ($\mathcal{C}_+ = 0.83$), so the vacuum-interior bound $f > 1/3$ of Sec. III.2 is a weak lower limit. For the first core model of Part I and $\rho_s = 13\,\rho_{\rm nuc}$, the absolute offset at $c_s^2 = 0.7$–1 becomes $\rho_* \approx 9.6$–$10.6\,\rho_{\rm nuc}$ (≈ 1.5–1.6 GeV/fm^3), larger than the vacuum-interior values of Sec. III.3. The dominant energy condition $|P_0| \le \sigma_0$ holds in all scanned models. In the tension cases the shell is a stretched membrane rather than a compressed layer, and the CSS parametrisation with $\rho_* > \rho_s$ describes it only formally.

Table III.2. Selected core models at $\mathcal{C}_+ = 0.785$. $f >$ is the lower bound $1 - w_0/\eta_{\rm crit}$ for a CSS shell. Values above 1 mean $\rho_* > \rho_s$, i.e. negative tangential pressure (tension).

$M\,[M_\odot]$	$\rho_c/\rho_{\rm nuc}$	α	w_0	$\eta_{\rm crit}$	$f >$
2.8	1	0.00	$+0.257$	0.661	0.61
2.8	2	0.15	$+0.182$	0.663	0.73
2.8	2	0.25	$+0.126$	0.673	0.81
2.8	1	0.50	$+0.044$	0.736	0.94
2.8	2	0.50	-0.155	0.710	1.22
3.2	4	0.50	-0.400	0.664	1.60

III.5 Interpretation and limits

The offset ρ_* is the model's quantitative handle on the Introduction of Part I: the hypothesis that the mass-generating structure is progressively switched off corresponds, in this language, to a large part of the energy density being stored in a pressure-free component. The result is that the shell works only if this component carries $\gtrsim 50\%$ of the shell energy. This is a consistency requirement, not evidence for the hypothesis.

Not established: (a) the fixed-thickness link $\eta_0 \leftrightarrow c_s^2$; (b) the core scan covers only two compactnesses (0.785, 0.83) and the static-core approximation of Part I; (c) the thin-shell approximation at the quoted ρ_s; (d) the literature comparison above; (e) any dynamics. Next: a finite-thickness layer with the CSS law, integrated with the anisotropic TOV equations, would decide whether the thin limit is representative.

Acknowledgments

Derivations and numerical checks were carried out with AI assistance (Claude, Anthropic) under the author's direction, who is responsible for the content. Numerical inputs: Part I, Table I.2; closed forms: Part II.

PART IV

Finite-Thickness Test of the Anisotropic Thin Shell: The Thin Shell as an Einstein-Cluster Limit, and Local Energy Conditions

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Abstract. Parts I–III rely on the thin-shell (Darmois–Israel) description. Here we test whether it is representative by replacing the shell with a layer of finite thickness l between a flat vacuum interior and a Schwarzschild exterior, and solving the anisotropic TOV equations inside it. We prove analytically that the equilibrium of Part II, $s_0 = 1 - y$, $\Pi_0 = (1-y)^2/(2y)$, $P_0/\sigma_0 = (1-y)/(4y)$ with $y = \sqrt{1-\mathcal{C}_+}$, is exactly the thin limit of a layer carrying tangential pressure only (an Einstein-cluster shell), whatever the density profile. Numerically, the mean ratio $\langle p_t\rangle/\langle\rho\rangle$ differs from the thin-shell value by 0.3% at $l/a = 0.1$ and 1.2% at $l/a = 0.2$, so the equilibrium input of Parts II–III is robust against thickness. The thin shell does, however, hide a local problem: for $p_r = 0$ the local ratio p_t/ρ at the outer surface equals $\mathcal{C}_+/[4(1-\mathcal{C}_+)]$, which violates the dominant energy condition (DEC) for $\mathcal{C}_+ > 0.8$. A small radial pressure (peak $p_r/\rho \lesssim 0.04$) removes this violation: with a one-parameter profile the local DEC holds up to $\mathcal{C}_+ \simeq 0.89$–$0.90$, which covers the whole stable, causal window $0.785 \le \mathcal{C}_+ \le 0.8437$ of Part II. Radial stability of the thick layer is not computed here.

IV.1 Model

Units $G = c = 1$. Take a layer $r_1 \le r \le a$, $l = a - r_1$, $\varepsilon = l/a$, with flat vacuum for $r < r_1$ and Schwarzschild vacuum of mass M for $r > a$. Inside the layer, the anisotropic TOV equations [1] read

$$m' = 4\pi r^2 \rho, \tag{IV.1}$$

$$p_r' = -\frac{(\rho + p_r)(m + 4\pi r^3 p_r)}{r(r-2m)} + \frac{2(p_t - p_r)}{r}. \tag{IV.2}$$

Note that this layer model differs from the core–shell configuration of Part I, where the interior fluid has $p_r(a) = \rho/3 \neq 0$ on the inner side of the shell; here the interior is vacuum and p_r vanishes on both faces, so the test applies to the vacuum-interior limit of Parts II–III, not directly to the configurations with a TOV core. Continuity of m and p_r with the vacuum regions requires $p_r(r_1) = p_r(a) = 0$. Equation (IV.2) is solved for p_t given a density profile and a radial pressure profile. The density is taken uniform, $\rho = \rho_s$ (as in Part I, Table I.2), and the radial pressure either vanishes or follows $p_r = 4A\rho_s x(1-x)$ with $x = (r - r_1)/l$. The surface

quantities are defined with proper length $d\ell = dr/\sqrt{1-2m/r}$,

$$\sigma = \int \rho\, d\ell, \qquad P = \int p_t\, d\ell, \qquad s = 4\pi a\sigma, \qquad \Pi = 8\pi a P. \tag{IV.3}$$

IV.2 The thin shell is an Einstein-cluster limit

Proposition 4. *Let $p_r \equiv 0$ in the layer. Then $p_t = \rho m/[2(r-2m)]$, and in the limit $\varepsilon \to 0$, for any density profile,*

$$s = 1-y, \qquad \Pi = \frac{(1-y)^2}{2y}, \qquad \frac{P}{\sigma} = \frac{1-y}{4y}, \qquad y = \sqrt{1-\mathcal{C}_+}. \tag{IV.4}$$

These are exactly the equilibrium values of Part II, Eq. (II.4).

Proof. Setting $p_r = 0$ in (IV.2) gives $p_t = \rho m/[2(r-2m)]$. In the thin limit $r \to a$, $dm = 4\pi a^2 \rho\, dr$ and $u = 2m/a \in [0, \mathcal{C}_+]$. Then

$$s = 4\pi a \int \frac{\rho\, dr}{\sqrt{1-u}} = \int_0^{\mathcal{C}_+} \frac{du}{2\sqrt{1-u}} = 1 - \sqrt{1-\mathcal{C}_+} = 1-y, \tag{IV.5}$$

$$\Pi = 8\pi a \int \frac{p_t\, dr}{\sqrt{1-u}} = \frac{1}{4}\int_0^{\mathcal{C}_+} \frac{u\, du}{(1-u)^{3/2}} = \frac{1}{4}\left[\frac{2}{y} + 2y - 4\right] = \frac{(1-y)^2}{2y}. \tag{IV.6}$$

Neither integral depends on the profile $\rho(r)$ because both are integrals over m. Dividing gives $P/\sigma = \Pi/(2s)$. □

This gives the equilibrium of Part II a concrete microscopic reading: a thin shell whose matter has negligible radial pressure and whose tangential pressure carries the weight against gravity. It also explains why the equilibrium ratio w_0 is a prediction of the model rather than a free parameter.

IV.3 Numerical results

The layer equations were evaluated on a grid of 4001 points (trapezoid rule). Validation against the proposition at $\varepsilon = 10^{-3}$: s and Π agree to $\sim 10^{-3}$ (the expected $\mathcal{O}(\varepsilon)$ correction) and P/σ agrees to five digits (e.g. 0.28916 for $\mathcal{C}_+ = 0.785$, equal to $(1-y)/(4y)$).

Table IV.1. Thickness dependence for tangential-only layers. The mean ratio is almost independent of l/a. The maximum local ratio is attained at the outer surface and equals $\mathcal{C}_+/[4(1-\mathcal{C}_+)]$ (analytic), independent of thickness. The last column is the uniform density needed for $M = 2.8\,M_\odot$ at that l/a.

$\mathcal{C}_+$	l/a	$\langle p_t\rangle/\langle\rho\rangle$	thin-shell w_0	max p_t/ρ ($p_r=0$)	$\rho_s/\rho_{\rm nuc}$ ($2.8\,M_\odot$)
0.785	0.05	0.2890	0.2892	0.913	29.5
0.785	0.10	0.2884	0.2892	0.913	15.5
0.785	0.20	0.2857	0.2892	0.913	8.6
0.820	0.10	0.3383	0.3393	1.139	17.7
0.844	0.10	0.3818	0.3830	1.353	19.3

Local energy conditions. Since $p_t/\rho = m/[2(r-2m)]$ is monotonic in m, its maximum is at the outer surface, $\mathcal{C}_+/[4(1-\mathcal{C}_+)]$, which exceeds 1 for $\mathcal{C}_+ > 0.8$. The thin-shell DEC of Part

I ($\sigma \geq |P|$) tests only the mean $P/\sigma = (1-y)/(4y)$, and is therefore weaker than the local condition. For tangential-only layers the local DEC (and local causality of the tangential stress) holds only for $\mathcal{C}_+ \leq 0.8$.

Effect of a small radial pressure. With $p_r = 4A\rho_s x(1-x)$ the term p_r' in (IV.2) is negative on the outer half and lowers p_t there. Minimising $\max(|p_t|, p_r)/\rho$ over A: for $\mathcal{C}_+ = 0.82$, $l/a = 0.1$ the best amplitude is $A \approx 0.03$ and the maximum ratio drops from 1.14 to 0.54; for $\mathcal{C}_+ = 0.844$, $l/a = 0.1$ from 1.35 to 0.65. The largest compactness for which the local DEC can be satisfied within this one-parameter family is $\mathcal{C}_+ \simeq 0.889$ ($l/a = 0.05$), 0.894 ($l/a = 0.1$), 0.900 ($l/a = 0.2$). This is a property of the chosen profile family and not a bound.

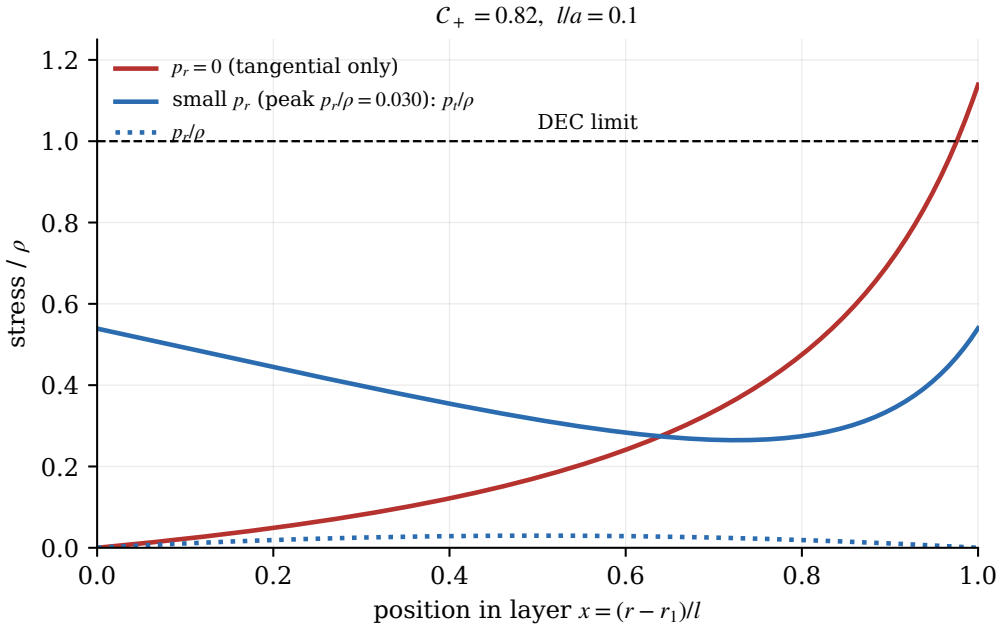


Figure IV.1. Local stresses across the layer at $\mathcal{C}_+ = 0.82$, $l/a = 0.1$. For $p_r = 0$ the tangential pressure exceeds the DEC limit near the outer surface; a small radial pressure (dotted) removes the excess.

IV.4 Consequences for Parts II and III

1. The equilibrium input of Parts II and III ($w_0 = (1-y)/(4y)$) is robust: thickness changes it by $\lesssim 0.3\%$ at $l/a = 0.1$. The value of w_0 itself refers to a vacuum interior; with a TOV core it is lower and depends on the core anisotropy (Part II, Sec. II.3.5, and Part III), which this layer model does not include. The closed forms $\eta_{\rm vac}$, $\Gamma_{\rm min}$, $\Gamma_{\rm max}$ and $\mathcal{C}_+^*$ therefore remain the relevant conditions for stability of the thin limit.

2. The thin-shell DEC window of Part I should be restated (a note to this effect is added to Sec. I.3 of Part I in this edition): for tangential-only matter the local DEC holds only for $\mathcal{C}_+ \leq 0.8$, whereas the stable, causal window of Part II extends to $\mathcal{C}_+^* = 0.8437$. In $0.8 < \mathcal{C}_+ \leq 0.8437$ the shell needs a small radial pressure of order $0.03\rho_s$ or less, which then also enters the dynamics.

3. The uniform-density requirement is consistent with Part I: $\rho_s \simeq 15.5\,\rho_{\rm nuc}$ at $l/a = 0.1$, $\mathcal{C}_+ = 0.785$ (Part I: 13–19 $\rho_{\rm nuc}$).

Not established. The radial stability of the thick layer is not analysed: it requires closures for both $p_r(\rho)$ and $p_t(\rho)$ and a radial-mode problem, and the identification of η_0 with a tangential sound speed (Parts II–III) is still an assumption. The profile families (uniform ρ, parabolic p_r) are choices, and the vacuum interior replaces the TOV core of Part I. The CSS offset of Part III has not been re-derived for the thick layer.

Acknowledgments

Derivations and numerical computations were carried out with AI assistance (Claude, Anthropic) under the author's direction, who is responsible for the content.

Bibliography

[1] M. Słowik, *Anisotropic Thin Shell Around a* $3\,M_{\odot}$ *Core: Static Equilibrium, Radial Stability, Vacuum Limit, and Thickness Constraints*, preprint (November 4, 2026). [Part I]

[2] M. Słowik, *Equation of State of the Anisotropic Thin Shell: Exact Stability Window, No-Go for Linear Closures, and an Exterior Echo-Delay Estimate*, preprint (November 5, 2026). [Part II]

[3] M. Słowik, *A Constant-Sound-Speed Realisation of the Stable Anisotropic Thin Shell and the Required Vacuum-Offset Energy*, preprint (November 2026). [Part III]

[4] M. Słowik, *Finite-Thickness Test of the Anisotropic Thin Shell: The Thin Shell as an Einstein-Cluster Limit, and Local Energy Conditions*, preprint (November 2026). [Part IV]

[5] J. R. Oppenheimer, G. M. Volkoff, On massive neutron cores, *Phys. Rev.* **55** (1939) 374.

[6] H. A. Buchdahl, General relativistic fluid spheres, *Phys. Rev.* **116** (1959) 1027.

[7] R. L. Bowers, E. P. T. Liang, Anisotropic spheres in general relativity, *Astrophys. J.* **188** (1974) 657.

[8] J. B. Hartle, Slowly rotating relativistic stars. I., *Astrophys. J.* **150** (1967) 1005.

[9] L. Herrera, N. O. Santos, Local anisotropy in self-gravitating systems, *Phys. Rep.* **286** (1997) 53.

[10] W. Israel, Singular hypersurfaces and thin shells in general relativity, *Il Nuovo Cimento B* **44** (1966) 1.

[11] E. Poisson, M. Visser, Thin-shell wormholes: linearization stability, *Phys. Rev. D* **52** (1995) 7318.

[12] V. Cardoso, P. Pani, Testing the nature of dark compact objects: a status report, *Living Rev. Relativ.* **22** (2019) 4.

[13] H. O. Silva, C. F. B. Macedo, E. Berti, L. C. B. Crispino, Slowly rotating anisotropic neutron stars in general relativity and scalar-tensor theory, *Class. Quantum Grav.* **32** (2015) 145008.